

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 06 · Graphs of Functions

Questions

Student book · questions with writing space

- 56 pages
- Past papers up to June 2026

Graphs of Functions

Your classified route

19
questions

165
marks

3
families

8
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Graphs of Functions

Families and variants

F01 • Polynomial graph grammar

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

V02 Reconstruct a cubic from graph data, then intersect another curve

V03 Horizontal-line intersections and geometric separation

F02 • Reciprocal graphs and intersection counts

V01 Shifted a/x graph with line-intersection discriminant

V02 Reciprocal-square symmetry and four-intersection parameter range

V03 Count real roots by comparing polynomial and reciprocal sketches

F03 • Curves, intersections, regions and models

V01 Line-quadratic intersections and region inequalities

V02 Compare contextual quadratic and linear models with a valid domain

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Polynomial graph grammar

WHAT THIS FAMILY ASKS

Use factorised polynomial structure, root multiplicity, intercepts, end behaviour and algebraic form to sketch, read or reconstruct cubic graphs.

VARIANT ROUTE

V01 Factorised cubic: intercepts, multiplicity, sign and linked algebra

V02 Reconstruct a cubic from graph data, then intersect another curve

V03 Horizontal-line intersections and geometric separation

5 Complete Classified questions

47 total marks

10.

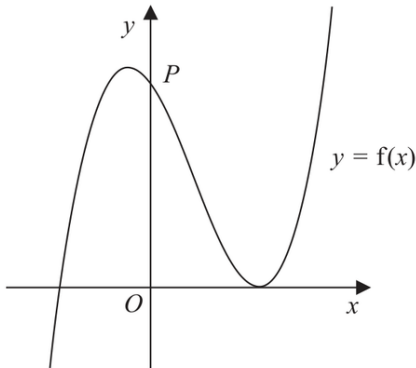


Figure 6

Figure 6 shows a sketch of part of the curve with equation $y = f(x)$, where

$$f(x) = (2x + 5)(x - 3)^2$$

(a) Deduce the values of x for which $f(x) \leq 0$

(2)


$$f(x) = (2x + 5)(x - 3)^2$$

(2)


$$f(x) = (2x + 5)(x - 3)^2$$

(b) Expand $f(x)$ to the form

where a, b, c and d are integers to be found.

(3)


$$f(x) = (2x + 5)(x - 3)^2$$

- (2)


$$f(x) = (2x + 5)(x - 3)^2$$

(d) (i) Find $g(x)$, giving your answer in a simplified factorised form.

(3)

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

(b) Hence or otherwise

(i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

(ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

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10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

(c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

(ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

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6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

In this question you must show all stages of your working.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found. **(3)**

6. In this question you must show all stages of your working.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

Figure 3 shows a sketch of part of the curve C_1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

(a) find $f(x)$

(3)

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The curve C_2 has equation $y = 1.2x(8 - x)$

(b) On Diagram 1 sketch a graph of the curve C_2

(2)

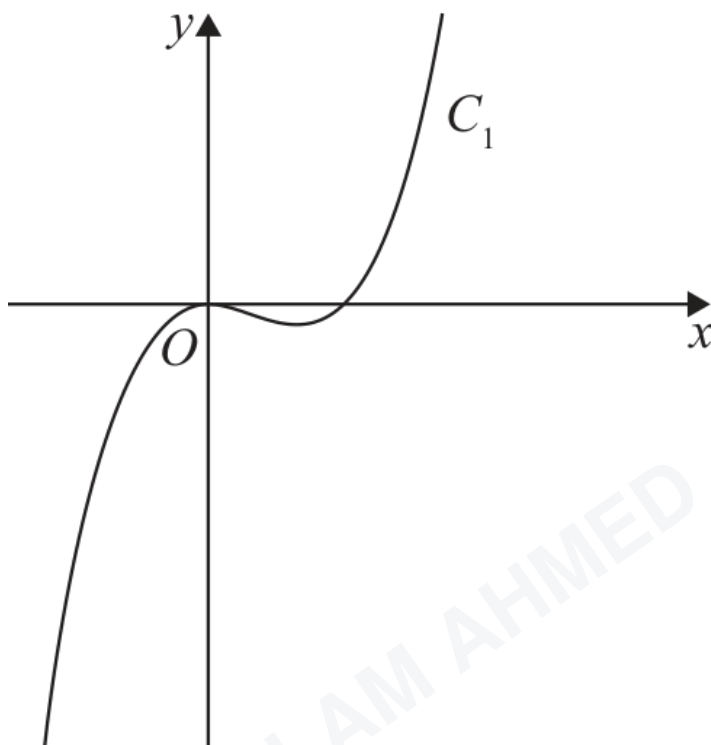


Diagram 1

Given that C_1

- has equation $y = f(x)$ where $f(x)$ is a cubic function
- touches the x -axis at the origin and cuts the x -axis at $x = 4$
- passes through the point $(10, 120)$

The curve C_2 has equation $y = 1.2x(8 - x)$

(c) Use algebra to find the coordinates of the points where C_1 and C_2 intersect. Show each stage of your working.

(5)

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

(a) Deduce the values of x for which $f(x) > 0$

(1)

Figure 4 shows a sketch of part of the curve C with equation $y = f(x)$, where

$$f(x) = (3x - 2)^2 (x - 4)$$

(b) Expand $f(x)$ to the form

$$ax^3 + bx^2 + cx + d$$

where a, b, c and d are integers to be found.

(3)


$$f(x) = (3x - 2)^2 (x - 4)$$

The line l cuts C again at points P and Q , also shown in Figure 4.

- (c) Using algebra and showing your working, find the length of line PQ . Write your answer in the form $k\sqrt{3}$, where k is a constant to be found.

(Solutions relying entirely on calculator technology are not acceptable.)

(5)

Reciprocal graphs and intersection counts

WHAT THIS FAMILY ASKS

Sketch translated reciprocal or reciprocal-square curves and connect their asymptotes, intercepts and symmetry with equations whose number of intersections is controlled algebraically or graphically.

VARIANT ROUTE

V01 Shifted a/x graph with line-intersection discriminant

V02 Reciprocal-square symmetry and four-intersection parameter range

V03 Count real roots by comparing polynomial and reciprocal sketches

11 Complete Classified questions

93 total marks

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

(a) Sketch a graph of C , stating the equation of the horizontal asymptote and the coordinates of the point of intersection with the x -axis.

(3)

6. The curve C has equation $y = \frac{4}{x} + k$, where k is a positive constant.

The line with equation $y = 10 - 2x$ is a tangent to C .

(b) Find the possible values for k .

(5)

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6. (a) Sketch the curve with equation

$$y = -\frac{k}{x} \quad k > 0 \quad x \neq 0$$

(2)

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(b) On a separate diagram, sketch the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

stating the coordinates of the point of intersection with the x -axis and, in terms of k , the equation of the horizontal asymptote.

(3)

(c) Find the range of possible values of k for which the curve with equation

$$y = -\frac{k}{x} + k \quad k > 0 \quad x \neq 0$$

does not touch or intersect the line with equation $y = 3x + 4$

(5)

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7. (a) Sketch the graph of the curve C with equation

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Show on your sketch

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

$$y = \frac{4}{x - k}$$

where k is a positive constant.

Given that the straight line with equation $y = 9 - x$ does not cross or touch C

(b) find the range of values of k .

(5)

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6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

(a) Sketch the curve C with equation

$$y = \frac{1}{2-x} \quad x \neq 2$$

State on your sketch

- the equation of the vertical asymptote
- the coordinates of the intersection of C with the y -axis

(3)

$$y = \frac{1}{2-x} \quad x \neq 2$$

The straight line l has equation $y = kx - 4$, where k is a constant.

Given that l cuts C at least once,

(b) (i) show that

$$k^2 - 5k + 4 \geq 0$$

(ii) find the range of possible values for k .

(6)

In this question you must show all stages of your working. Solutions relying entirely on calculator technology are not acceptable.

$$y = \frac{2}{x} - k$$

(a) Sketch the graph of C .

- the coordinates of any points of intersection of C with the coordinate axes
- the equation of the horizontal asymptote to C

(3)

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The curve C has equation

$$y = \frac{2}{x} - k$$

where k is a **positive** constant.

The line l has equation $y = -kx - 6$

Given that l intersects C at 2 distinct points,

(b) find the range of possible values of k .

(5)

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6. (a) Sketch the graph of the curve C with equation

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

On your sketch show

- the coordinates of any points where C cuts the coordinate axes
- the equation of the vertical asymptote to C

(4)

$$y = \frac{4k}{x - 2k}$$

where k is a positive constant.

The straight line l has equation

$$y = 6 - 2x$$

Given that there is at least one point of intersection between l and C ,

(b) find the range of possible values of k .

(5)

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

- (a) Sketch a graph of C .

Show on your sketch the coordinates of the point where C crosses a coordinate axis.

(3)

10. The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

The line l has equation $x + y = 12$

Given that l is a tangent to C ,

(b) show that

$$k^2 - 15k + 36 = 0$$

(4)

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The curve C has equation

$$y = \frac{3k}{x - 2k} \quad x \neq 2k$$

where k is a positive constant.

The line l has equation $x + y = 12$

$$k^2 - 15k + 36 = 0$$

The line l meets C at a point P .

Given that P lies below the x -axis,

(c) find the coordinates of P .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

(a) Sketch the graph of C .

On your sketch

- show the coordinates of any points of intersection with the coordinate axes
- state clearly the equations of any asymptotes

(4)

10. The curve C has equation

$$y = \frac{1}{x^2} - 9$$

The curve D has equation $y = kx^2$ where k is a constant.

Given that C meets D at 4 distinct points,

(b) find the range of possible values for k .

(5)

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8. The curve C_1 has equation

$$y = x(4 - x^2)$$

- (a) Sketch the graph of C_1 showing the coordinates of any points of intersection with the coordinate axes.

(3)

8. The curve C_1 has equation

$$y = x(4 - x^2)$$

The curve C_2 has equation $y = \frac{A}{x}$ where A is a constant.

(b) Show that the x coordinates of the points of intersection of C_1 and C_2 satisfy the equation

$$x^4 - 4x^2 + A = 0 \quad (1)$$

(c) Hence find the range of possible values of A for which C_1 meets C_2 at 4 distinct points.

(3)

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6. (a) Given that k is a positive constant such that $0 < k < 4$ sketch, on **separate axes**, the graphs of

(i) $y = (2x - k)(x + 4)^2$

$$\text{(ii) } y = \frac{k}{x^2}$$

showing the coordinates of any points where the graphs cross or meet the coordinate axes, leaving coordinates in terms of k , where appropriate.

(5)

Given that k is a positive constant such that $0 < k < 4$

(i) $y = (2x - k)(x + 4)^2$

(ii) $y = \frac{k}{x^2}$

(b) State, with a reason, the number of roots of the equation

$$(2x - k)(x + 4)^2 = \frac{k}{x^2}$$

(1)

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4.

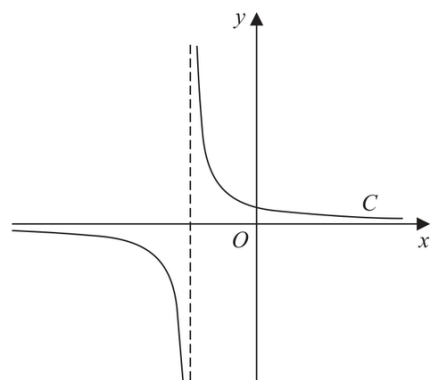


Figure 1

Figure 1 shows a sketch of part of the curve C with equation $y = \frac{1}{x+2}$

(a) State the equation of the asymptote of C that is parallel to the y -axis.

(1)

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(b) Factorise fully $x^3 + 4x^2 + 4x$

(2)

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A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(c) On Diagram 1, add a sketch of the curve with equation

$$y = x^3 + 4x^2 + 4x$$

On your sketch, state clearly the coordinates of each point where this curve cuts or meets the coordinate axes.

(3)

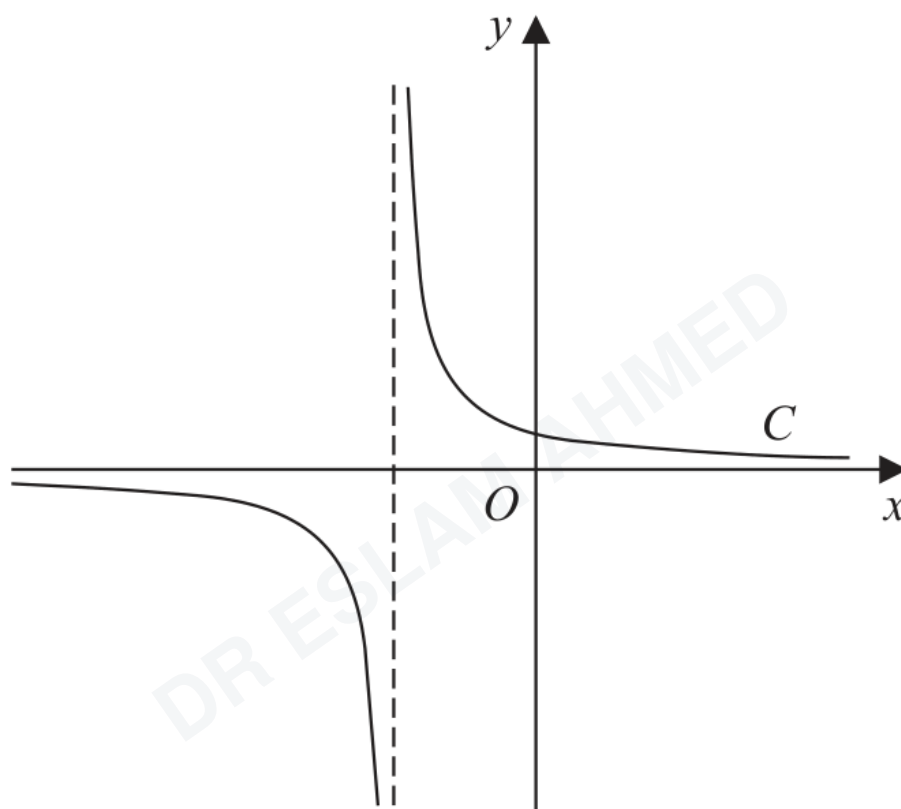


Diagram 1

$$y = \frac{1}{x+2}$$

$$y = x^3 + 4x^2 + 4x$$

(d) Hence state the number of real solutions of the equation

$$(x+2)(x^3 + 4x^2 + 4x) = 1$$

giving a reason for your answer.

(1)

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Curves, intersections, regions and models

WHAT THIS FAMILY ASKS

Use a graph as a geometric interpretation of simultaneous equations, inequalities and contextual function models.

VARIANT ROUTE

V01 Line-quadratic intersections and region inequalities

V02 Compare contextual quadratic and linear models with a valid domain

3 Complete Classified questions

25 total marks

Solutions relying on calculator technology are not acceptable.

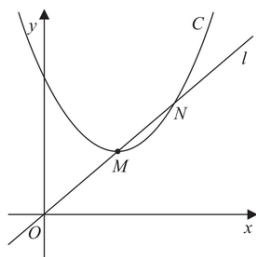


Figure 2 shows a sketch of the curve C with equation $y = x^2 - 5x + 13$

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

(3)

$$y = x^2 - 5x + 13$$

The point M is the minimum point of C .

The straight line l passes through the origin O and intersects C at the points M and N as shown.

Find, showing your working,

(b) the coordinates of N .

(5)

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$$y = x^2 - 5x + 13$$

The point M is the minimum point of C .

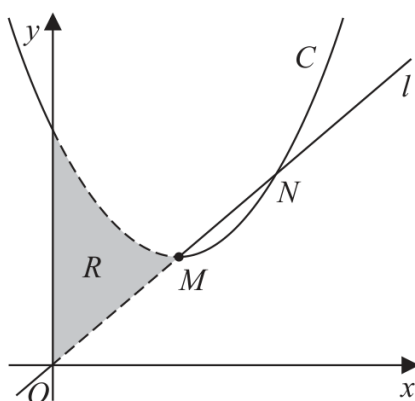


Figure 3

Figure 3 shows the curve C and the line l . The finite region R , shown shaded in Figure 3, is bounded by C , l and the y -axis.

(c) Use inequalities to define the region R .

(2)

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

The line l intersects the curve C at the points P and Q as shown in Figure 1.

- (4)

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

The line l intersects the curve C at the points P and Q as shown in Figure 1.

(b) Use inequalities to define the region R .

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

$$P_s = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

Figure 2 shows a graph of both models.

(a) Find the difference between the share value of **company A** and the share value of **company B** at the point monitoring began.

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of **company A**, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_s = -1.6t + 44.2 \quad t \geq 0$$

Figure 2 shows a graph of both models.

(1)

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of company A , in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

$$P_A = 53 - 0,4(t - 8)^2 \quad t \geq 0$$

The share value P_B of **company B**, in millions of pounds, is modelled by the equation

$$P_n = -1.6t + 44.2 \quad t \geq 0$$

Figure 2 shows a graph of both models.

- (c) Find, using algebra and showing your working, the times during this 15-year period when the share value of **company A** was greater than the share value of **company B**. (4)

5.

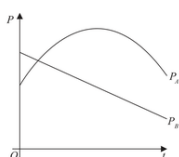


Figure 2

The share value of two companies, company *A* and company *B*, has been monitored over a 15-year period.

The share value P_A of company *A*, in millions of pounds, is modelled by the equation

$$P_A = 53 - 0.4(t - 8)^2 \quad t \geq 0$$

where t is the number of years after monitoring began.

The share value P_B of company *B*, in millions of pounds, is modelled by the equation

$$P_B = -1.6t + 44.2 \quad t \geq 0$$

where t is the number of years after monitoring began.

Figure 2 shows a graph of both models.

- (d) Explain why the model for **company A** should not be used to predict its share value when $t = 20$

(1)

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